

Mathematical Reflections — S705

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"All constructive problems are trivial."

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Prove that there are infinitely many triples (a, b, c) of distinct positive integers such that $a^3 + b^3 + c^3 - 3abc$ is a perfect square.

Solution

We make use of the well-known identity

$$a^3 + b^3 + c^3 - 3abc = \frac{1}{2}(a + b + c) [(a - b)^2 + (b - c)^2 + (c - a)^2]$$

Let $c < b < a$ be in arithmetic progression with common difference $d > 1$, where d is a positive integer. We have

$$\begin{aligned} a^3 + b^3 + c^3 - 3abc &= \frac{1}{2}(c + 3d) [d^2 + d^2 + 4d^2] \\ &= 3d^2(c + 3d) \end{aligned}$$

Now let $c = 3d^2 - 3d = 3d(d - 1)$ and we have $a^3 + b^3 + c^3 - 3abc = (3d^2)^2$